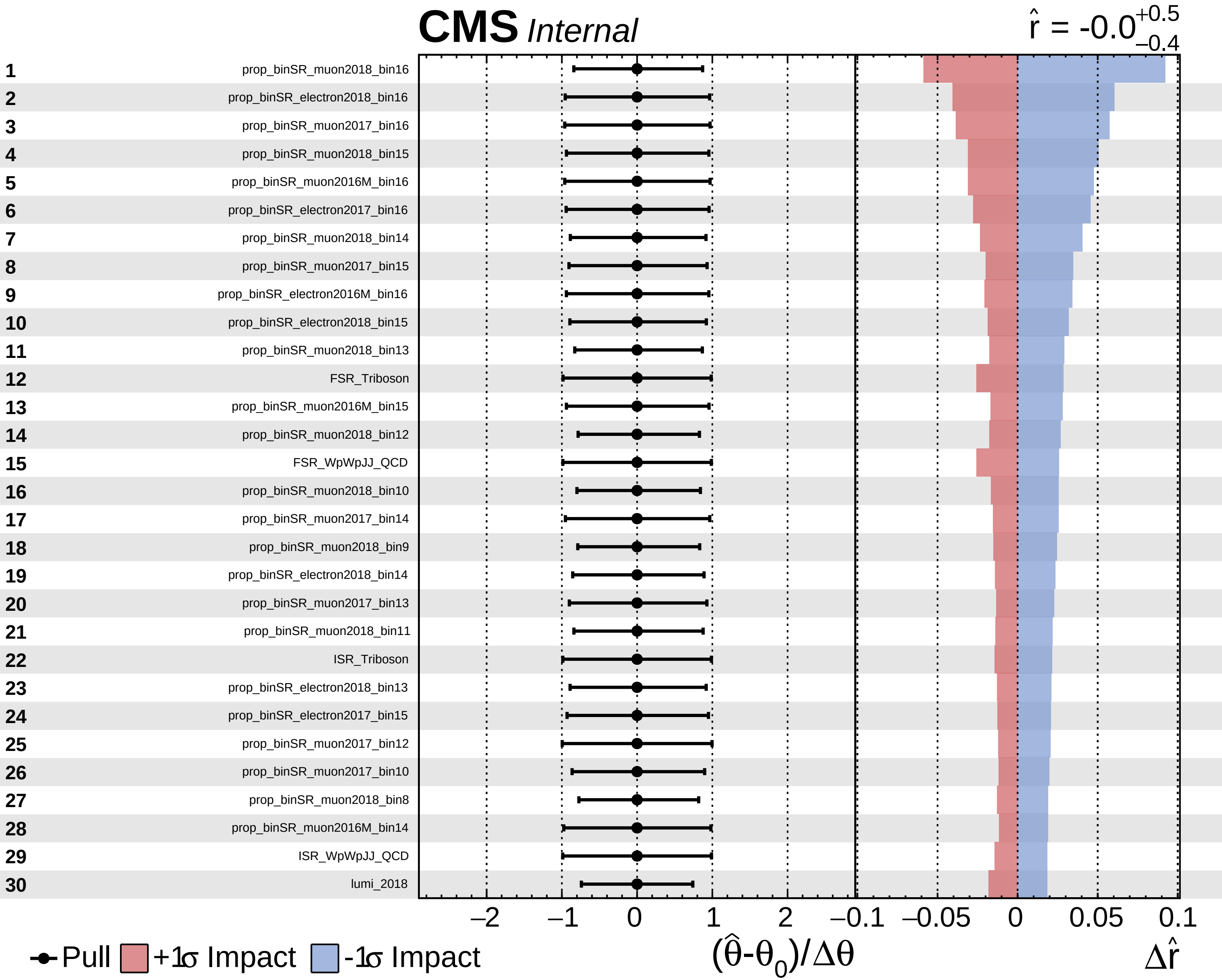
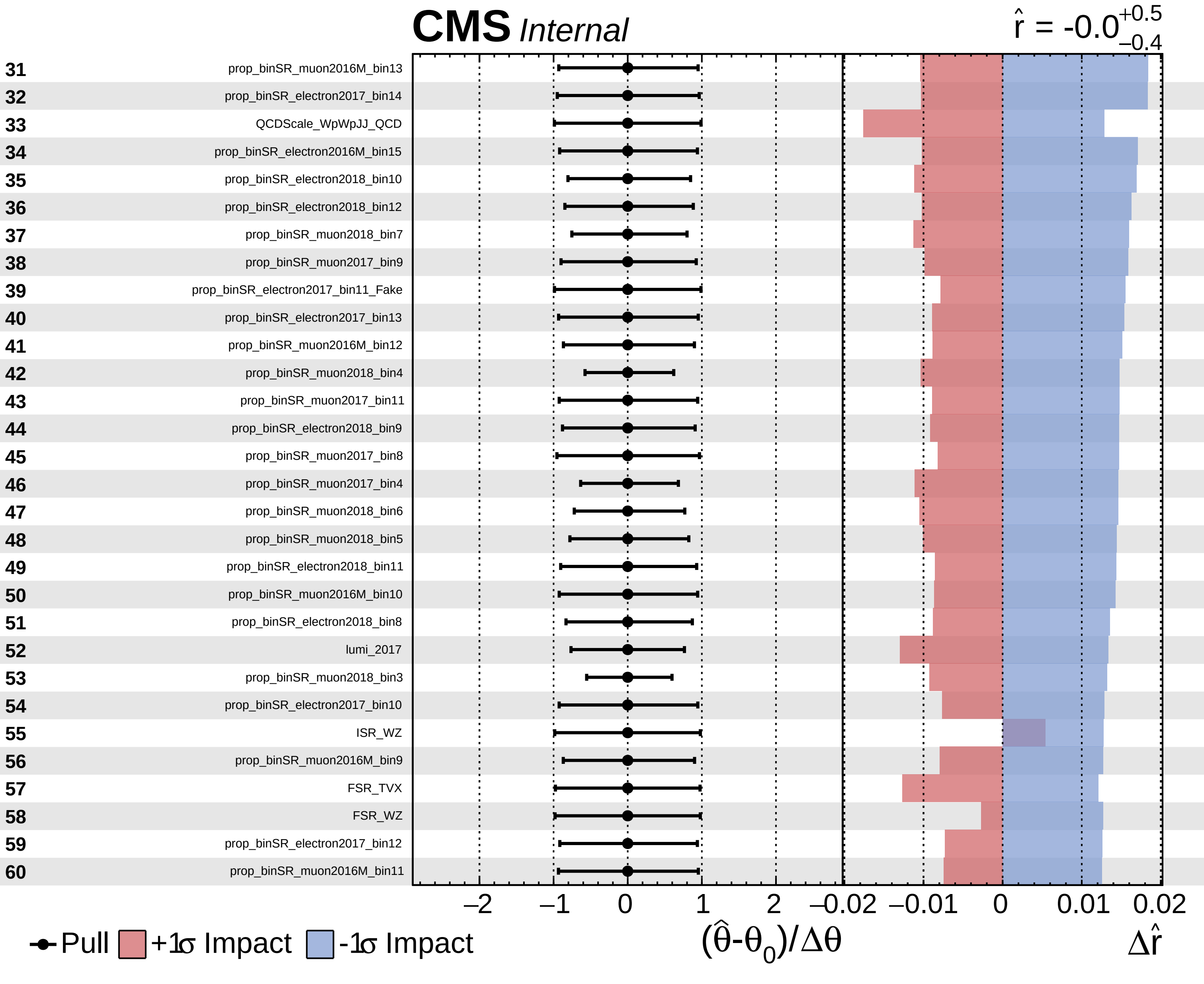
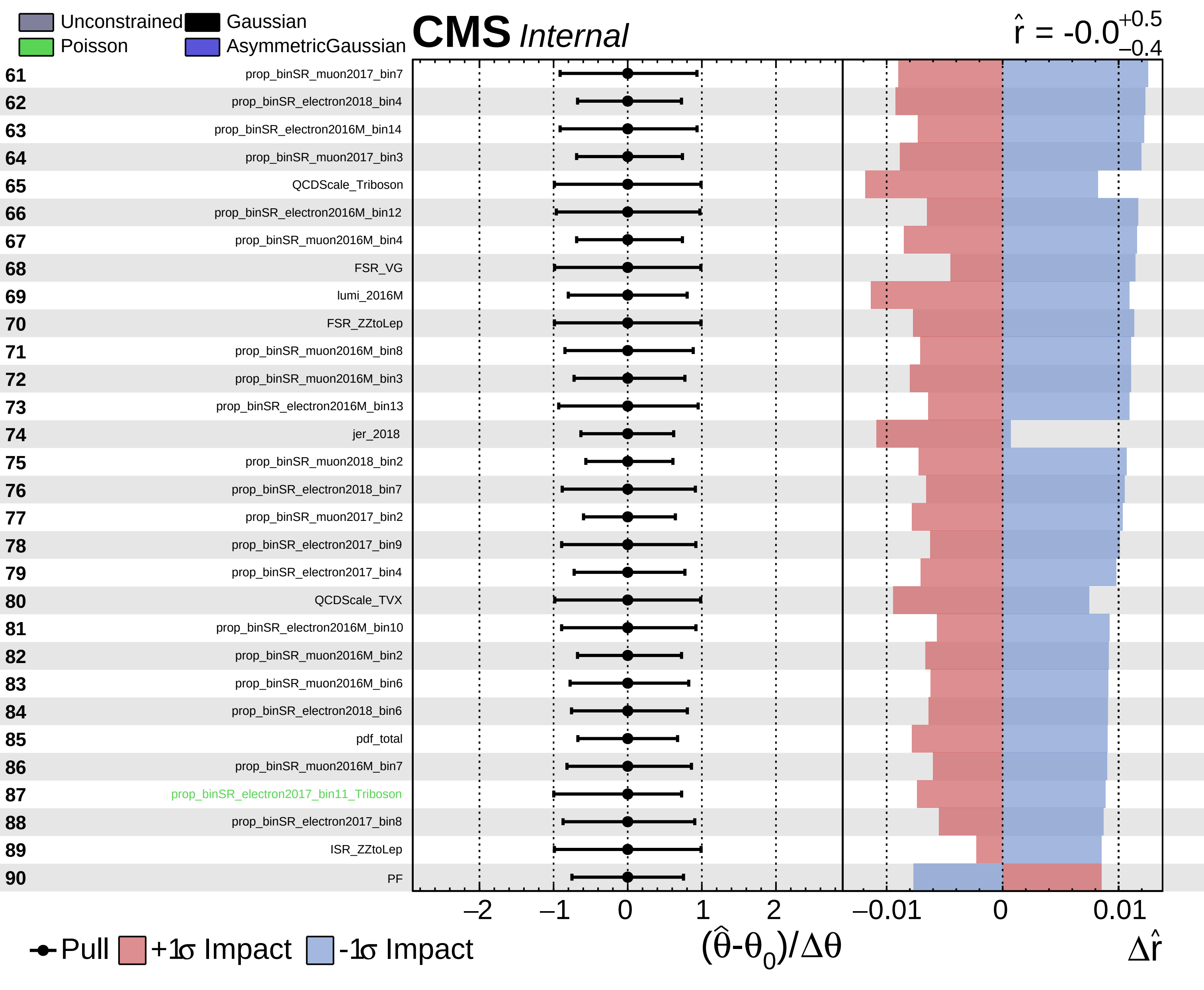


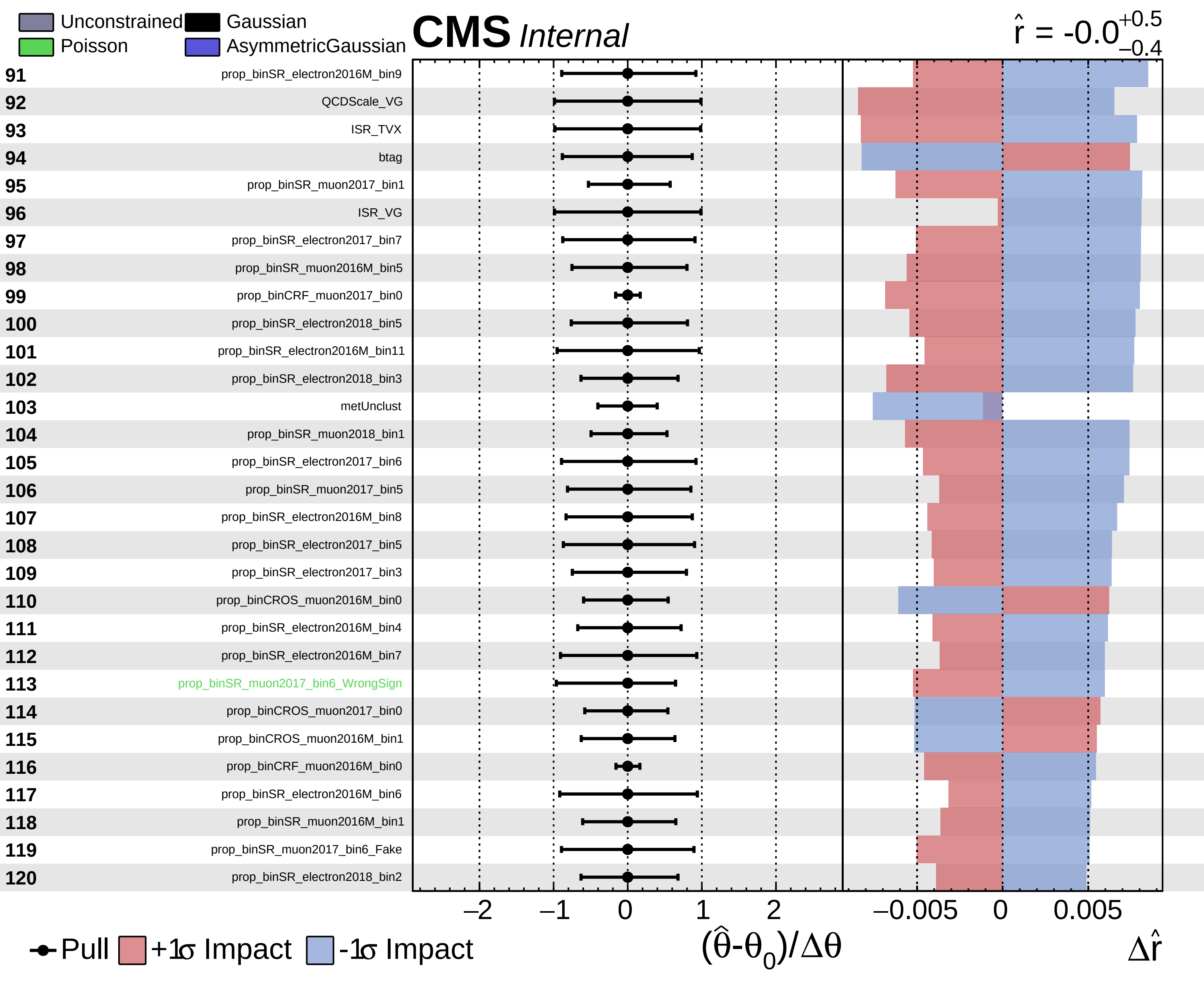
CMS Internal
$\hat{r} = -0.0^{+0.5}_{-0.4}$
1 prop_binSR_muon2018_bin16
2 prop_binSR_electron2018_bin16
3 prop_binSR_muon2017_bin16
4 prop_binSR_muon2018_bin15
5 prop_binSR_muon2016M_bin16
6 prop_binSR_electron2017_bin16
7 prop_binSR_muon2018_bin14
8 prop_binSR_muon2017_bin15
9 prop_binSR_electron2016M_bin16
10 prop_binSR_electron2018_bin15
11 prop_binSR_muon2018_bin13
12 FSR_Triboson
13 prop_binSR_muon2016M_bin15
14 prop_binSR_muon2018_bin12
15 FSR_WpWpJJ_QCD
16 prop_binSR_muon2018_bin10
17 prop_binSR_muon2017_bin14
18 prop_binSR_muon2018_bin9
19 prop_binSR_electron2018_bin14
20 prop_binSR_muon2017_bin13
21 prop_binSR_muon2018_bin11
22 ISR_Triboson
23 prop_binSR_electron2018_bin13
24 prop_binSR_electron2017_bin15
25 prop_binSR_muon2017_bin12
26 prop_binSR_muon2017_bin10
27 prop_binSR_muon2018_bin8
28 prop_binSR_muon2016M_bin14
29 ISR_WpWpJJ_QCD
30 lumi_2018
−2 −1 0 1 2
−0.1 −0.05 0 0.05 0.1
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



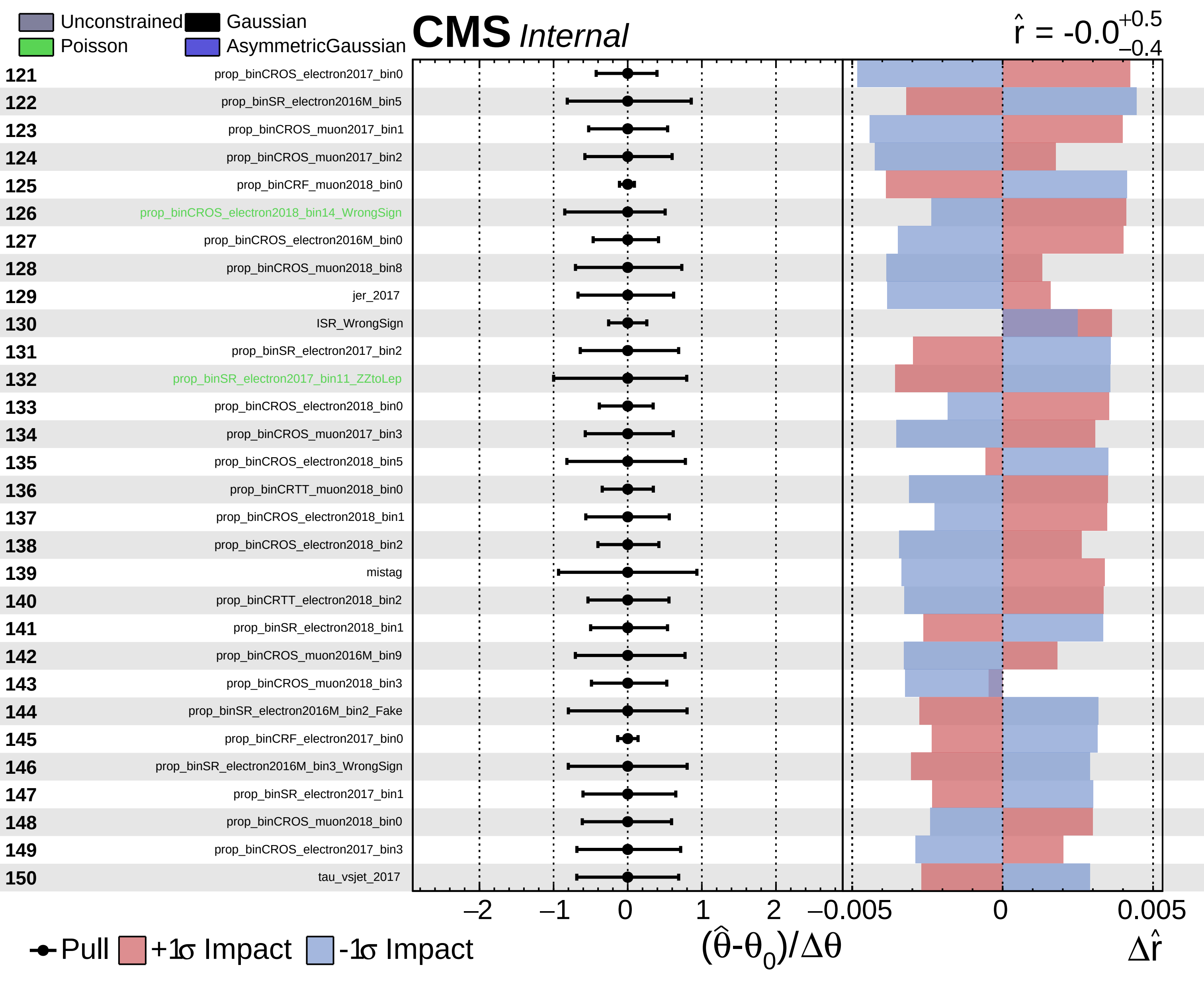
CMS Internal
$\hat{r} = -0.0^{+0.5}_{-0.4}$
31 prop_binSR_muon2016M_bin13
32 prop_binSR_electron2017_bin14
33 QCDScale_WpWpJJ_QCD
34 prop_binSR_electron2016M_bin15
35 prop_binSR_electron2018_bin10
36 prop_binSR_electron2018_bin12
37 prop_binSR_muon2018_bin7
38 prop_binSR_muon2017_bin9
39 prop_binSR_electron2017_bin11_Fake
40 prop_binSR_electron2017_bin13
41 prop_binSR_muon2016M_bin12
42 prop_binSR_muon2018_bin4
43 prop_binSR_muon2017_bin11
44 prop_binSR_electron2018_bin9
45 prop_binSR_muon2017_bin8
46 prop_binSR_muon2017_bin4
47 prop_binSR_muon2018_bin6
48 prop_binSR_muon2018_bin5
49 prop_binSR_electron2018_bin11
50 prop_binSR_muon2016M_bin10
51 prop_binSR_electron2018_bin8
52 lumi_2017
53 prop_binSR_muon2018_bin3
54 prop_binSR_electron2017_bin10
55 ISR_WZ
56 prop_binSR_muon2016M_bin9
57 FSR_TVX
58 FSR_WZ
59 prop_binSR_electron2017_bin12
60 prop_binSR_muon2016M_bin11
−2 −1 0 1 2
−0.02 −0.01 0 0.01 0.02
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



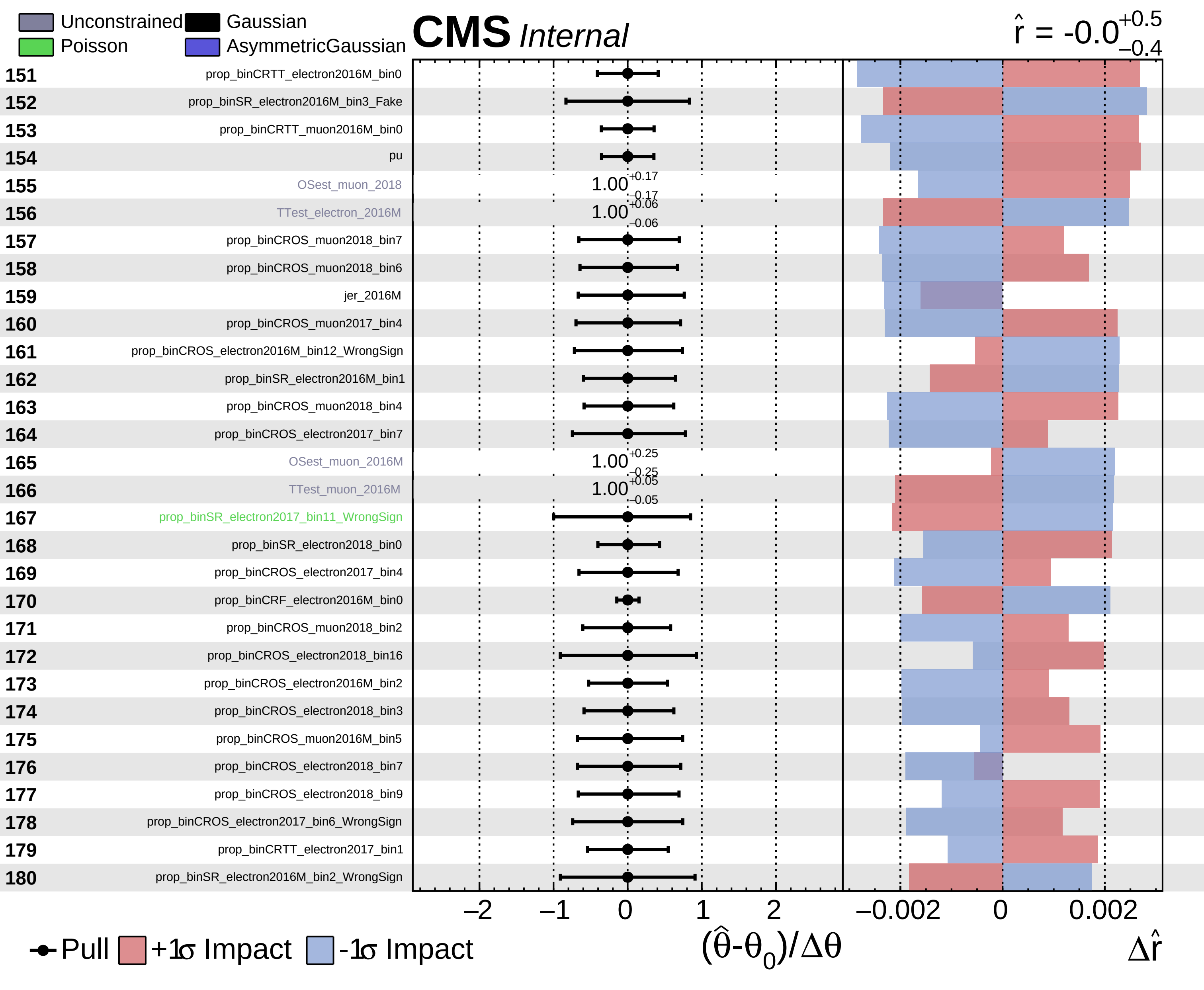
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.0^{+0.5}_{-0.4}$
61 prop_binSR_muon2017_bin7
62 prop_binSR_electron2018_bin4
63 prop_binSR_electron2016M_bin14
64 prop_binSR_muon2017_bin3
65 QCDScale_Triboson
66 prop_binSR_electron2016M_bin12
67 prop_binSR_muon2016M_bin4
68 FSR_VG
69 lumi_2016M
70 FSR_ZZtoLep
71 prop_binSR_muon2016M_bin8
72 prop_binSR_muon2016M_bin3
73 prop_binSR_electron2016M_bin13
74 jer_2018
75 prop_binSR_muon2018_bin2
76 prop_binSR_electron2018_bin7
77 prop_binSR_muon2017_bin2
78 prop_binSR_electron2017_bin9
79 prop_binSR_electron2017_bin4
80 QCDScale_TVX
81 prop_binSR_electron2016M_bin10
82 prop_binSR_muon2016M_bin2
83 prop_binSR_muon2016M_bin6
84 prop_binSR_electron2018_bin6
85 pdf_total
86 prop_binSR_muon2016M_bin7
87 prop_binSR_electron2017_bin11_Triboson
88 prop_binSR_electron2017_bin8
89 ISR_ZZtoLep
90 PF
−2
−1
0
1
2
−0.01
0
0.01
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



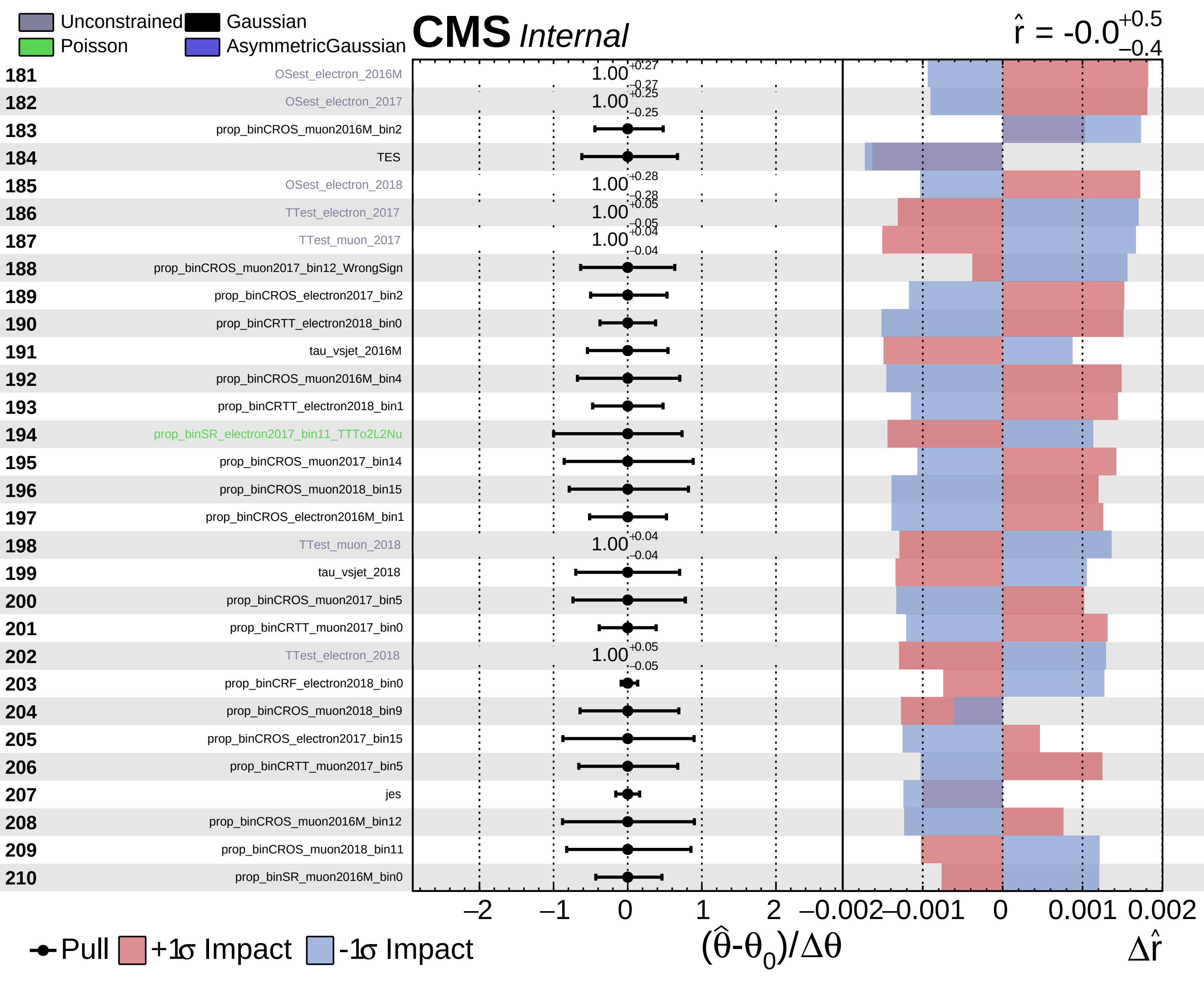
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.0^{+0.5}_{-0.4}$
91 prop_binSR_electron2016M_bin9
92 QCDScale_VG
93 ISR_TVX
94 btag
95 prop_binSR_muon2017_bin1
96 ISR_VG
97 prop_binSR_electron2017_bin7
98 prop_binSR_muon2016M_bin5
99 prop_binCRF_muon2017_bin0
100 prop_binSR_electron2018_bin5
101 prop_binSR_electron2016M_bin11
102 prop_binSR_electron2018_bin3
103 metUnclust
104 prop_binSR_muon2018_bin1
105 prop_binSR_electron2017_bin6
106 prop_binSR_muon2017_bin5
107 prop_binSR_electron2016M_bin8
108 prop_binSR_electron2017_bin5
109 prop_binSR_electron2017_bin3
110 prop_binCROS_muon2016M_bin0
111 prop_binSR_electron2016M_bin4
112 prop_binSR_electron2016M_bin7
113 prop_binSR_muon2017_bin6_WrongSign
114 prop_binCROS_muon2017_bin0
115 prop_binCROS_muon2016M_bin1
116 prop_binCRF_muon2016M_bin0
117 prop_binSR_electron2016M_bin6
118 prop_binSR_muon2016M_bin1
119 prop_binSR_muon2017_bin6_Fake
120 prop_binSR_electron2018_bin2
−2 −1 0 1 2
−0.005 0 0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



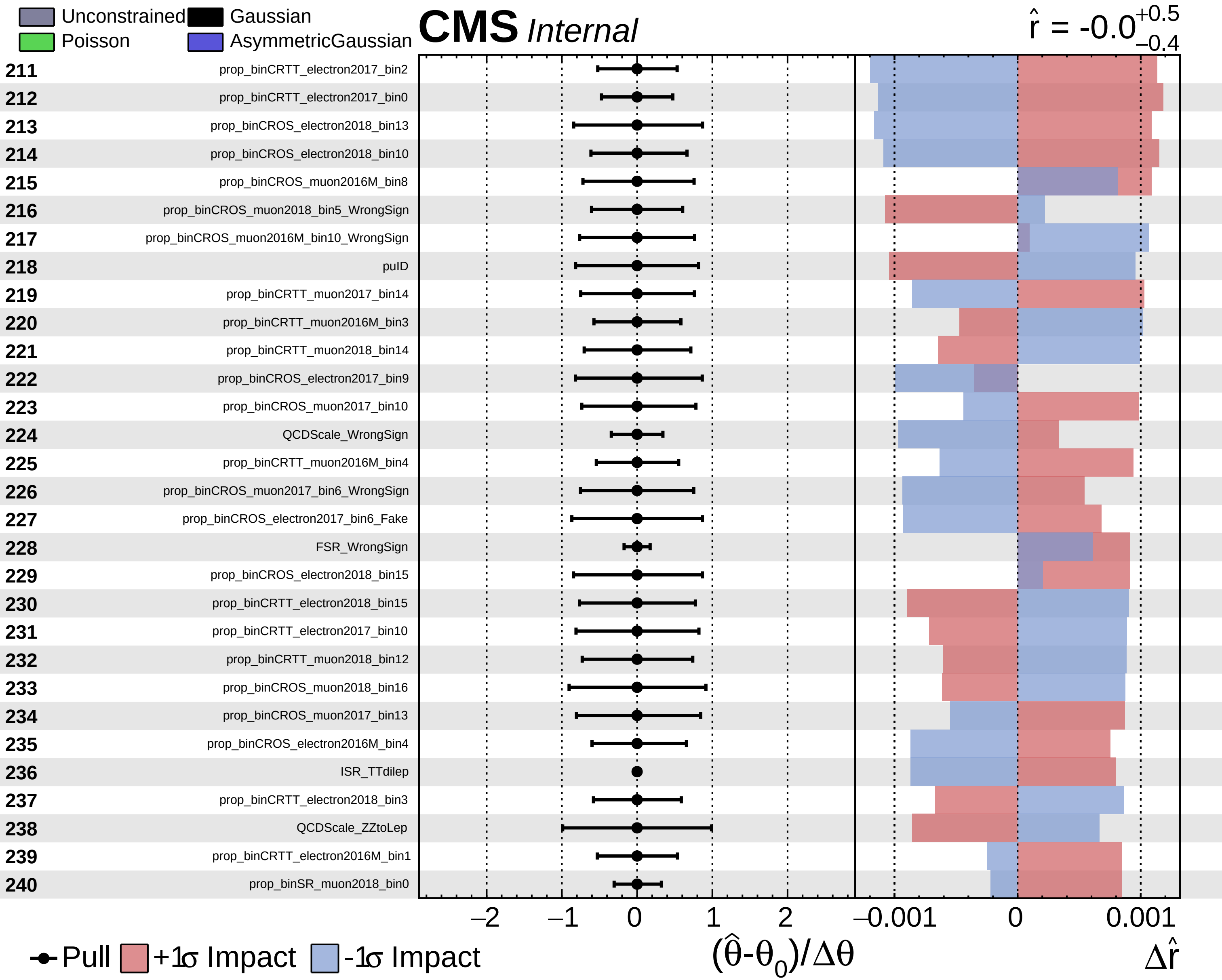
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.0^{+0.5}_{-0.4}$
121 prop_binCROS_electron2017_bin0
122 prop_binSR_electron2016M_bin5
123 prop_binCROS_muon2017_bin1
124 prop_binCROS_muon2017_bin2
125 prop_binCRF_muon2018_bin0
126 prop_binCROS_electron2018_bin14_WrongSign
127 prop_binCROS_electron2016M_bin0
128 prop_binCROS_muon2018_bin8
129 jer_2017
130 ISR_WrongSign
131 prop_binSR_electron2017_bin2
132 prop_binSR_electron2017_bin11_ZZtoLep
133 prop_binCROS_electron2018_bin0
134 prop_binCROS_muon2017_bin3
135 prop_binCROS_electron2018_bin5
136 prop_binCRTT_muon2018_bin0
137 prop_binCROS_electron2018_bin1
138 prop_binCROS_electron2018_bin2
139 mistag
140 prop_binCRTT_electron2018_bin2
141 prop_binSR_electron2018_bin1
142 prop_binCROS_muon2016M_bin9
143 prop_binCROS_muon2018_bin3
144 prop_binSR_electron2016M_bin2_Fake
145 prop_binCRF_electron2017_bin0
146 prop_binSR_electron2016M_bin3_WrongSign
147 prop_binSR_electron2017_bin1
148 prop_binCROS_muon2018_bin0
149 prop_binCROS_electron2017_bin3
150 tau_vsjet_2017
−2
−1
0
1
2
−0.005
0
0.005
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

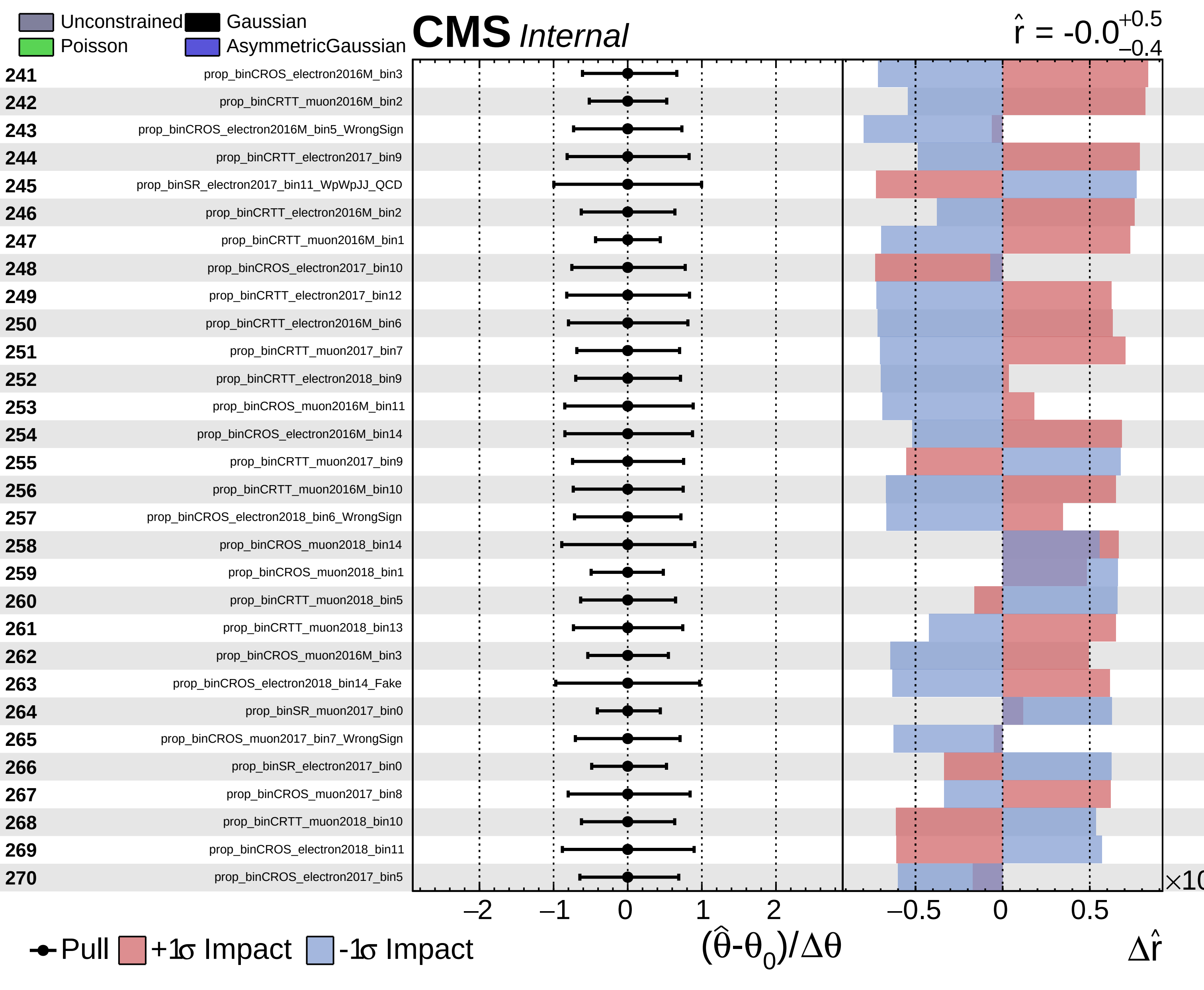


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.0^{+0.5}_{-0.4}$
151 prop_binCRTT_electron2016M_bin0
152 prop_binSR_electron2016M_bin3_Fake
153 prop_binCRTT_muon2016M_bin0
154 pu
155 OSest_muon_2018 1.00 +0.17 −0.17
156 TTest_electron_2016M 1.00 +0.06 −0.06
157 prop_binCROS_muon2018_bin7
158 prop_binCROS_muon2018_bin6
159 jer_2016M
160 prop_binCROS_muon2017_bin4
161 prop_binCROS_electron2016M_bin12_WrongSign
162 prop_binSR_electron2016M_bin1
163 prop_binCROS_muon2018_bin4
164 prop_binCROS_electron2017_bin7
165 OSest_muon_2016M 1.00 +0.25 −0.25
166 TTest_muon_2016M 1.00 +0.05 −0.05
167 prop_binSR_electron2017_bin11_WrongSign
168 prop_binSR_electron2018_bin0
169 prop_binCROS_electron2017_bin4
170 prop_binCRF_electron2016M_bin0
171 prop_binCROS_muon2018_bin2
172 prop_binCROS_electron2018_bin16
173 prop_binCROS_electron2016M_bin2
174 prop_binCROS_electron2018_bin3
175 prop_binCROS_muon2016M_bin5
176 prop_binCROS_electron2018_bin7
177 prop_binCROS_electron2018_bin9
178 prop_binCROS_electron2017_bin6_WrongSign
179 prop_binCRTT_electron2017_bin1
180 prop_binSR_electron2016M_bin2_WrongSign
−2 −1 0 1 2
−0.002 0 0.002
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

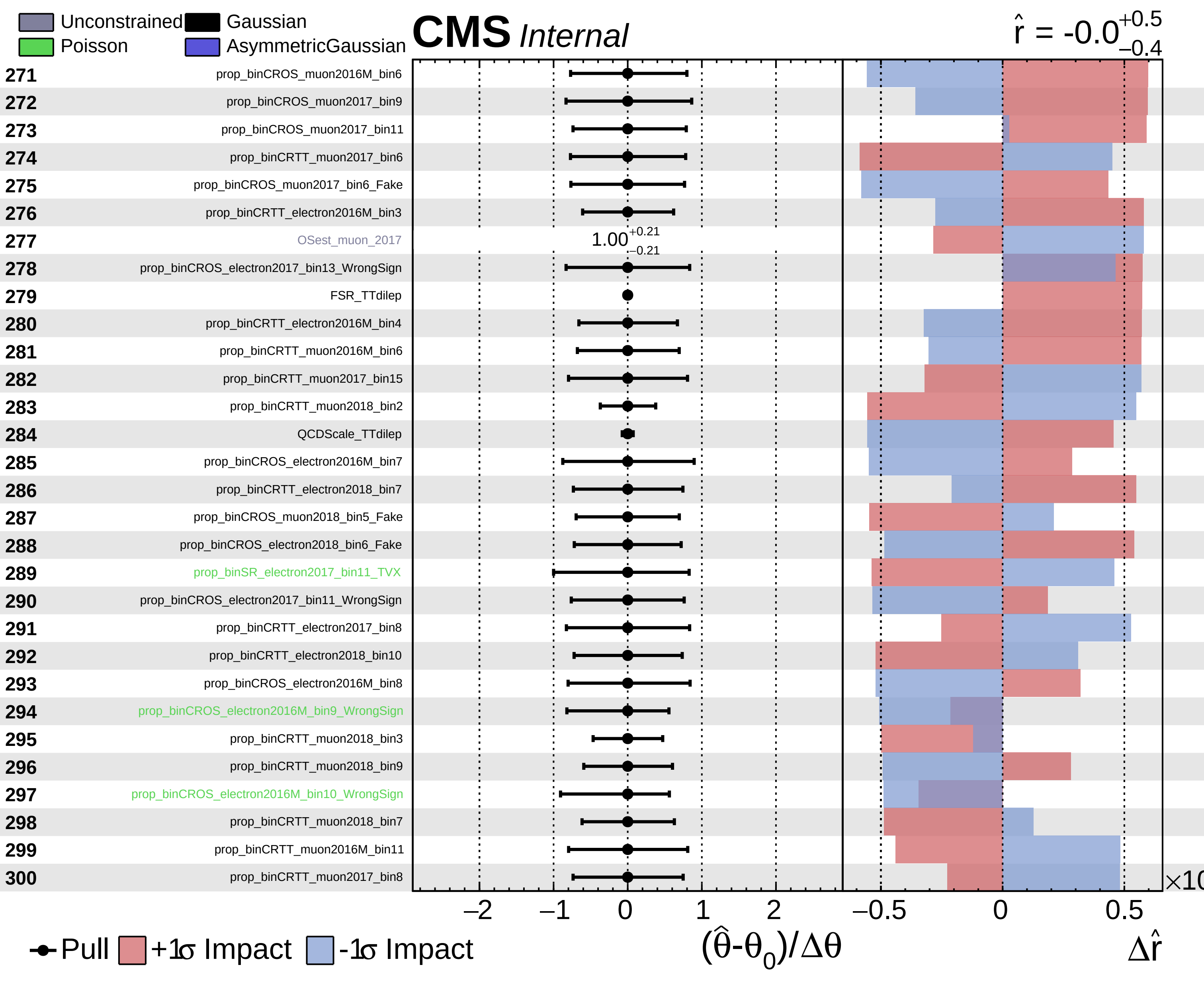


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = -0.0 +0.5 −0.4
181 OSest_electron_2016M 1.00 +0.27 −0.27
182 OSest_electron_2017 1.00 +0.25 −0.25
183 prop_binCROS_muon2016M_bin2
184 TES
185 OSest_electron_2018 1.00 +0.28 −0.28
186 TTest_electron_2017 1.00 +0.05 −0.05
187 TTest_muon_2017 1.00 +0.04 −0.04
188 prop_binCROS_muon2017_bin12_WrongSign
189 prop_binCROS_electron2017_bin2
190 prop_binCRTT_electron2018_bin0
191 tau_vsjet_2016M
192 prop_binCROS_muon2016M_bin4
193 prop_binCRTT_electron2018_bin1
194 prop_binSR_electron2017_bin11_TTTo2L2Nu
195 prop_binCROS_muon2017_bin14
196 prop_binCROS_muon2018_bin15
197 prop_binCROS_electron2016M_bin1
198 TTest_muon_2018 1.00 +0.04 −0.04
199 tau_vsjet_2018
200 prop_binCROS_muon2017_bin5
201 prop_binCRTT_muon2017_bin0
202 TTest_electron_2018 1.00 +0.05 −0.05
203 prop_binCRF_electron2018_bin0
204 prop_binCROS_muon2018_bin9
205 prop_binCROS_electron2017_bin15
206 prop_binCRTT_muon2017_bin5
207 jes
208 prop_binCROS_muon2016M_bin12
209 prop_binCROS_muon2018_bin11
210 prop_binSR_muon2016M_bin0
−2 −1 0 1 2
−0.002 −0.001 0 0.001 0.002
Pull
+1σ Impact
-1σ Impact
(θ̂-θ0)/Δθ
Δr̂

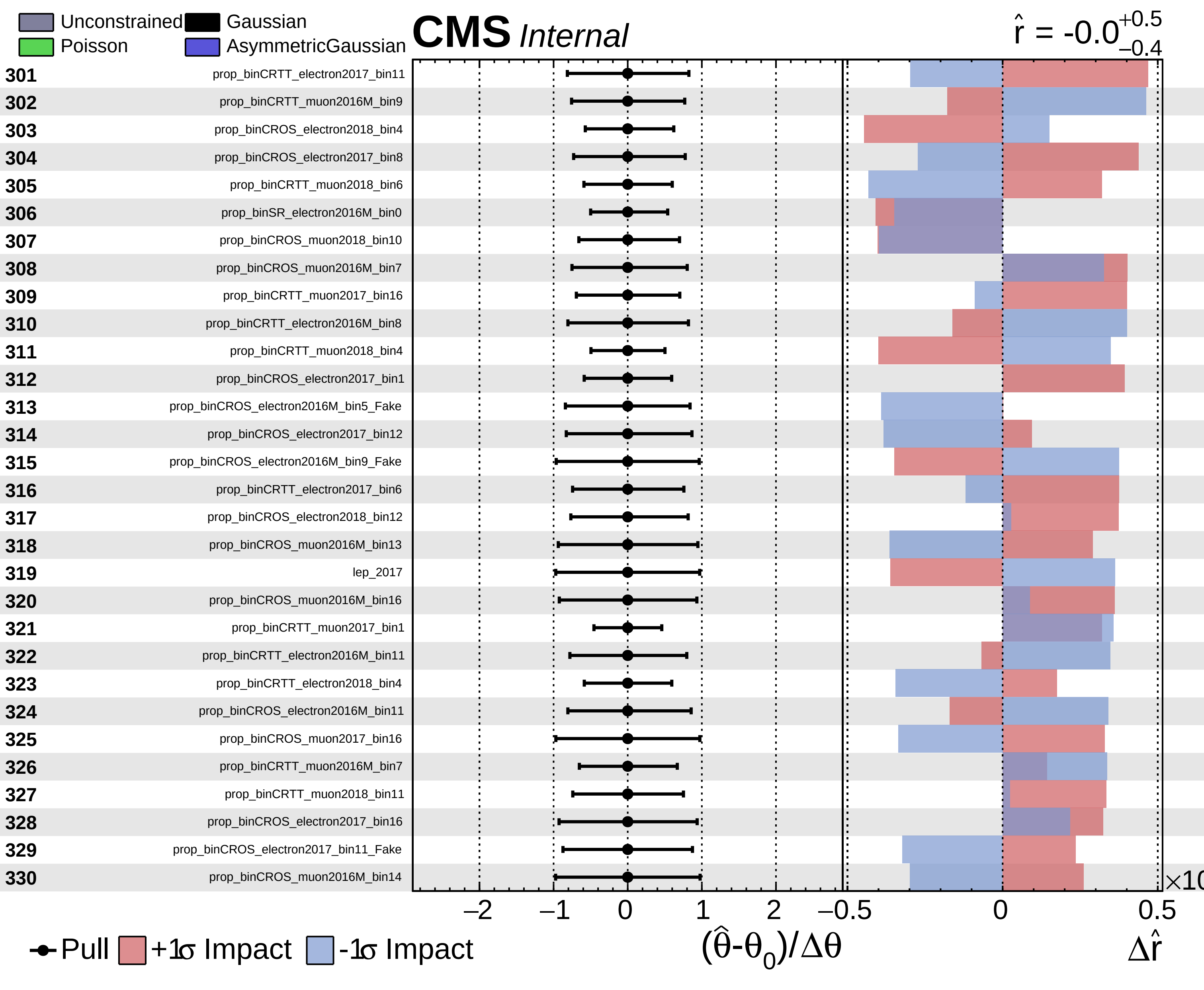

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.0^{+0.5}_{-0.4}$
211 prop_binCRTT_electron2017_bin2
212 prop_binCRTT_electron2017_bin0
213 prop_binCROS_electron2018_bin13
214 prop_binCROS_electron2018_bin10
215 prop_binCROS_muon2016M_bin8
216 prop_binCROS_muon2018_bin5_WrongSign
217 prop_binCROS_muon2016M_bin10_WrongSign
218 puID
219 prop_binCRTT_muon2017_bin14
220 prop_binCRTT_muon2016M_bin3
221 prop_binCRTT_muon2018_bin14
222 prop_binCROS_electron2017_bin9
223 prop_binCROS_muon2017_bin10
224 QCDScale_WrongSign
225 prop_binCRTT_muon2016M_bin4
226 prop_binCROS_muon2017_bin6_WrongSign
227 prop_binCROS_electron2017_bin6_Fake
228 FSR_WrongSign
229 prop_binCROS_electron2018_bin15
230 prop_binCRTT_electron2018_bin15
231 prop_binCRTT_electron2017_bin10
232 prop_binCRTT_muon2018_bin12
233 prop_binCROS_muon2018_bin16
234 prop_binCROS_muon2017_bin13
235 prop_binCROS_electron2016M_bin4
236 ISR_TTdilep
237 prop_binCRTT_electron2018_bin3
238 QCDScale_ZZtoLep
239 prop_binCRTT_electron2016M_bin1
240 prop_binSR_muon2018_bin0
−2
−1
0
1
2
−0.001
0
0.001
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

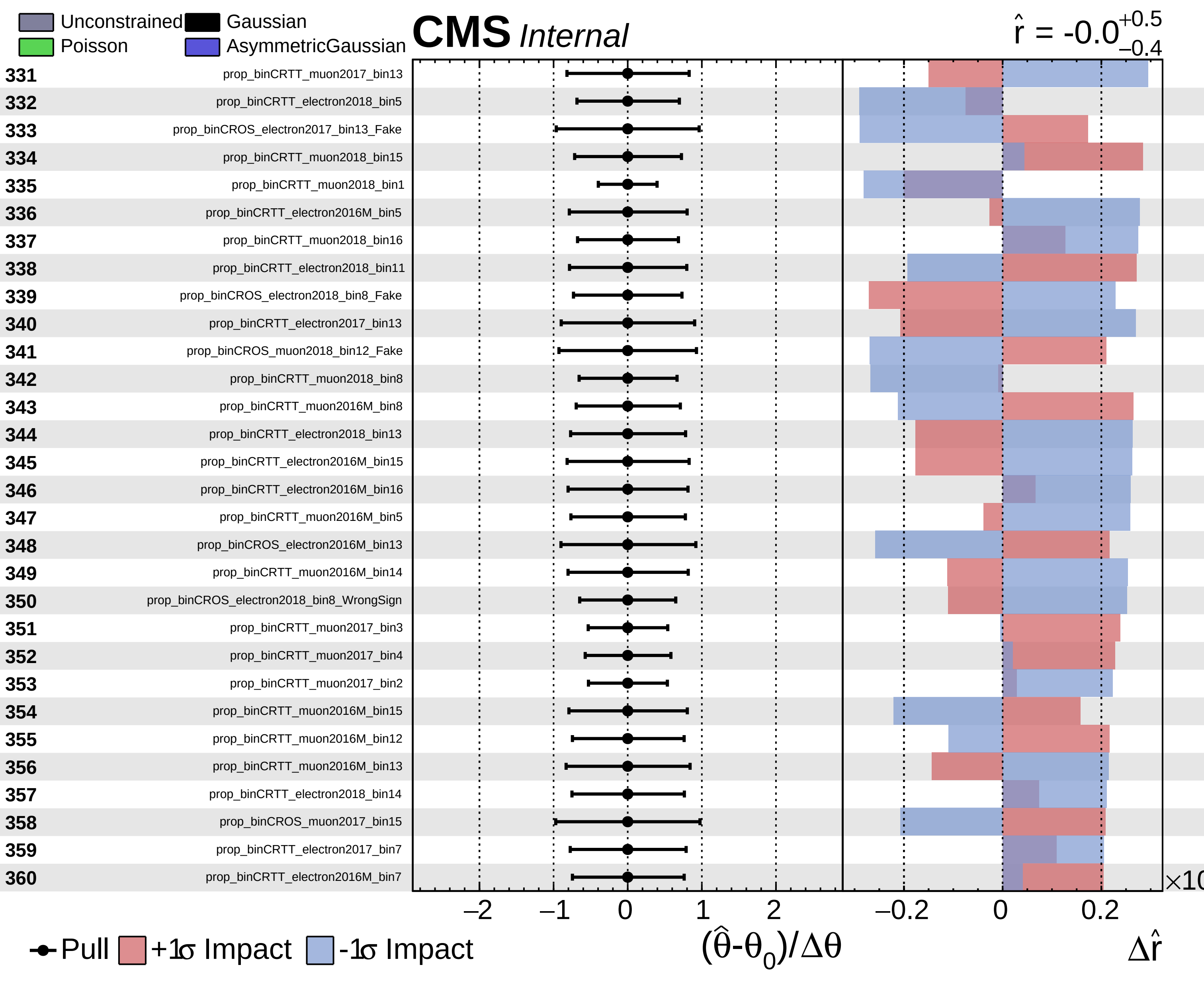


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
r̂ = -0.0 +0.5 −0.4
241 prop_binCROS_electron2016M_bin3
242 prop_binCRTT_muon2016M_bin2
243 prop_binCROS_electron2016M_bin5_WrongSign
244 prop_binCRTT_electron2017_bin9
245 prop_binSR_electron2017_bin11_WpWpJJ_QCD
246 prop_binCRTT_electron2016M_bin2
247 prop_binCRTT_muon2016M_bin1
248 prop_binCROS_electron2017_bin10
249 prop_binCRTT_electron2017_bin12
250 prop_binCRTT_electron2016M_bin6
251 prop_binCRTT_muon2017_bin7
252 prop_binCRTT_electron2018_bin9
253 prop_binCROS_muon2016M_bin11
254 prop_binCROS_electron2016M_bin14
255 prop_binCRTT_muon2017_bin9
256 prop_binCRTT_muon2016M_bin10
257 prop_binCROS_electron2018_bin6_WrongSign
258 prop_binCROS_muon2018_bin14
259 prop_binCROS_muon2018_bin1
260 prop_binCRTT_muon2018_bin5
261 prop_binCRTT_muon2018_bin13
262 prop_binCROS_muon2016M_bin3
263 prop_binCROS_electron2018_bin14_Fake
264 prop_binSR_muon2017_bin0
265 prop_binCROS_muon2017_bin7_WrongSign
266 prop_binSR_electron2017_bin0
267 prop_binCROS_muon2017_bin8
268 prop_binCRTT_muon2018_bin10
269 prop_binCROS_electron2018_bin11
270 prop_binCROS_electron2017_bin5
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull +1σ Impact -1σ Impact
(θ̂-θ0)/Δθ
Δr̂

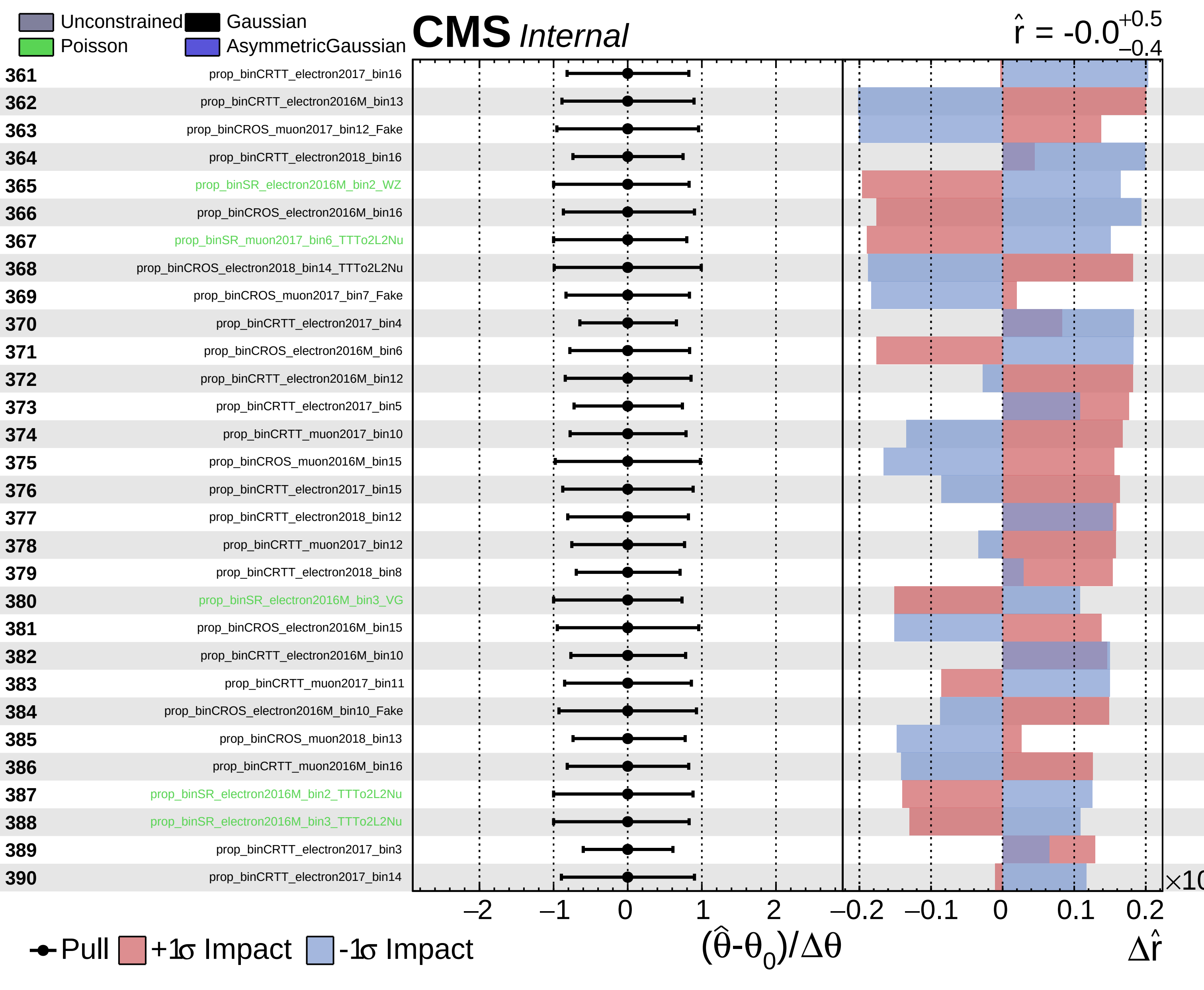


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.0^{+0.5}_{-0.4}$
271 prop_binCROS_muon2016M_bin6
272 prop_binCROS_muon2017_bin9
273 prop_binCROS_muon2017_bin11
274 prop_binCRTT_muon2017_bin6
275 prop_binCROS_muon2017_bin6_Fake
276 prop_binCRTT_electron2016M_bin3
277 OSest_muon_2017
$1.00^{+0.21}_{-0.21}$
278 prop_binCROS_electron2017_bin13_WrongSign
279 FSR_TTdilep
280 prop_binCRTT_electron2016M_bin4
281 prop_binCRTT_muon2016M_bin6
282 prop_binCRTT_muon2017_bin15
283 prop_binCRTT_muon2018_bin2
284 QCDScale_TTdilep
285 prop_binCROS_electron2016M_bin7
286 prop_binCRTT_electron2018_bin7
287 prop_binCROS_muon2018_bin5_Fake
288 prop_binCROS_electron2018_bin6_Fake
289 prop_binSR_electron2017_bin11_TVX
290 prop_binCROS_electron2017_bin11_WrongSign
291 prop_binCRTT_electron2017_bin8
292 prop_binCRTT_electron2018_bin10
293 prop_binCROS_electron2016M_bin8
294 prop_binCROS_electron2016M_bin9_WrongSign
295 prop_binCRTT_muon2018_bin3
296 prop_binCRTT_muon2018_bin9
297 prop_binCROS_electron2016M_bin10_WrongSign
298 prop_binCRTT_muon2018_bin7
299 prop_binCRTT_muon2016M_bin11
300 prop_binCRTT_muon2017_bin8
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

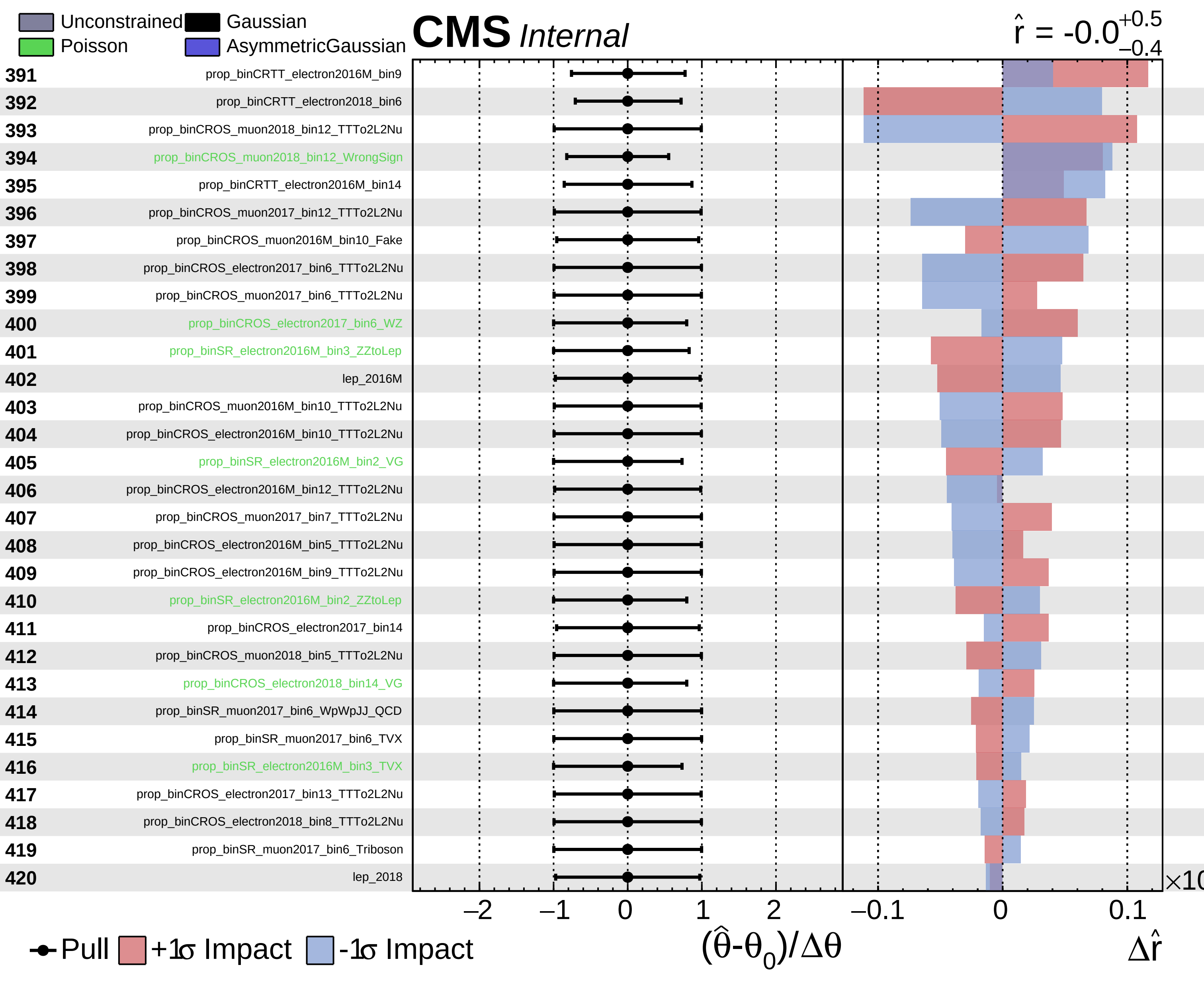

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.0^{+0.5}_{-0.4}$
301 prop_binCRTT_electron2017_bin11
302 prop_binCRTT_muon2016M_bin9
303 prop_binCROS_electron2018_bin4
304 prop_binCROS_electron2017_bin8
305 prop_binCRTT_muon2018_bin6
306 prop_binSR_electron2016M_bin0
307 prop_binCROS_muon2018_bin10
308 prop_binCROS_muon2016M_bin7
309 prop_binCRTT_muon2017_bin16
310 prop_binCRTT_electron2016M_bin8
311 prop_binCRTT_muon2018_bin4
312 prop_binCROS_electron2017_bin1
313 prop_binCROS_electron2016M_bin5_Fake
314 prop_binCROS_electron2017_bin12
315 prop_binCROS_electron2016M_bin9_Fake
316 prop_binCRTT_electron2017_bin6
317 prop_binCROS_electron2018_bin12
318 prop_binCROS_muon2016M_bin13
319 lep_2017
320 prop_binCROS_muon2016M_bin16
321 prop_binCRTT_muon2017_bin1
322 prop_binCRTT_electron2016M_bin11
323 prop_binCRTT_electron2018_bin4
324 prop_binCROS_electron2016M_bin11
325 prop_binCROS_muon2017_bin16
326 prop_binCRTT_muon2016M_bin7
327 prop_binCRTT_muon2018_bin11
328 prop_binCROS_electron2017_bin16
329 prop_binCROS_electron2017_bin11_Fake
330 prop_binCROS_muon2016M_bin14
−2 −1 0 1 2
−0.5 0 0.5
×10
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

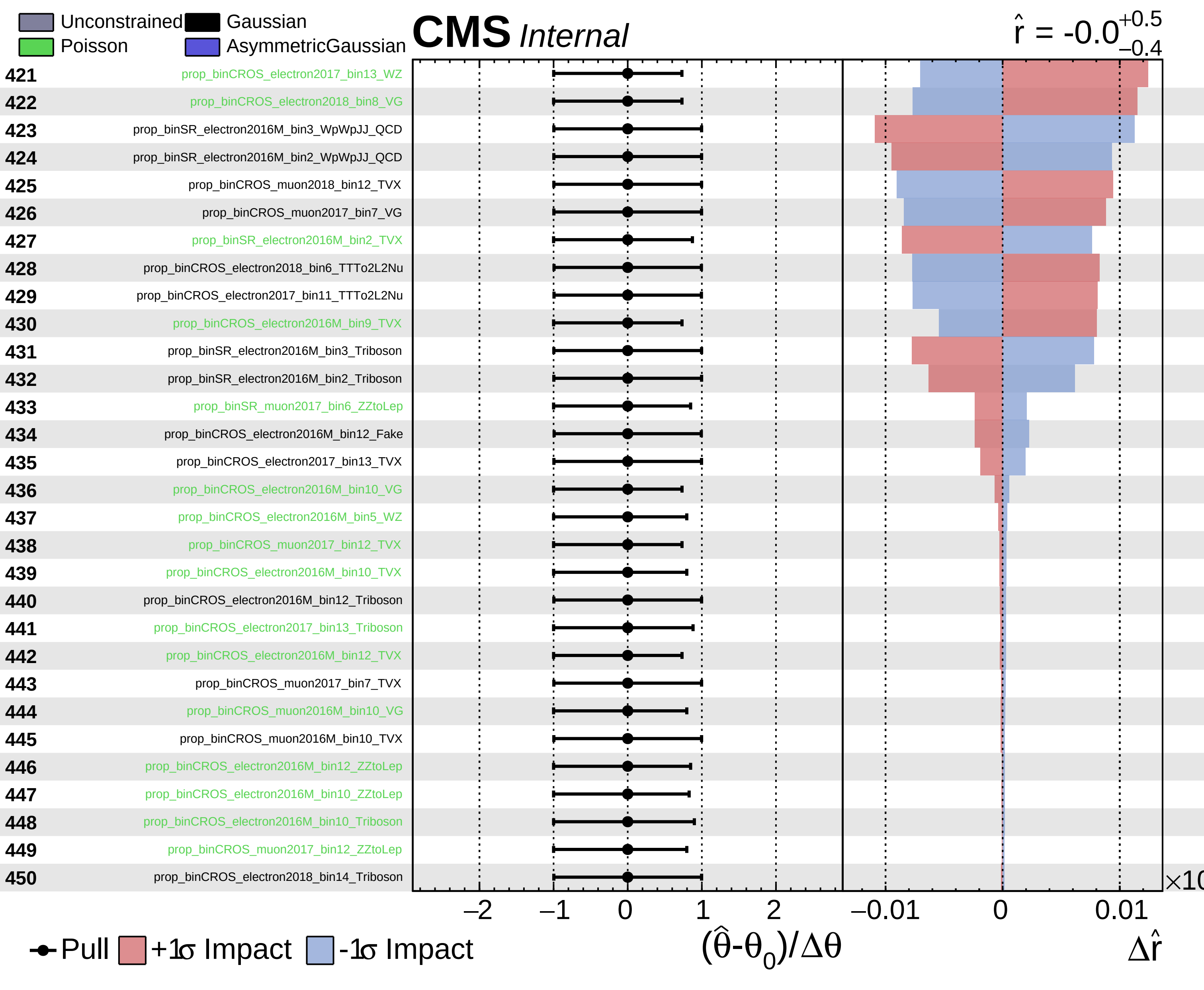


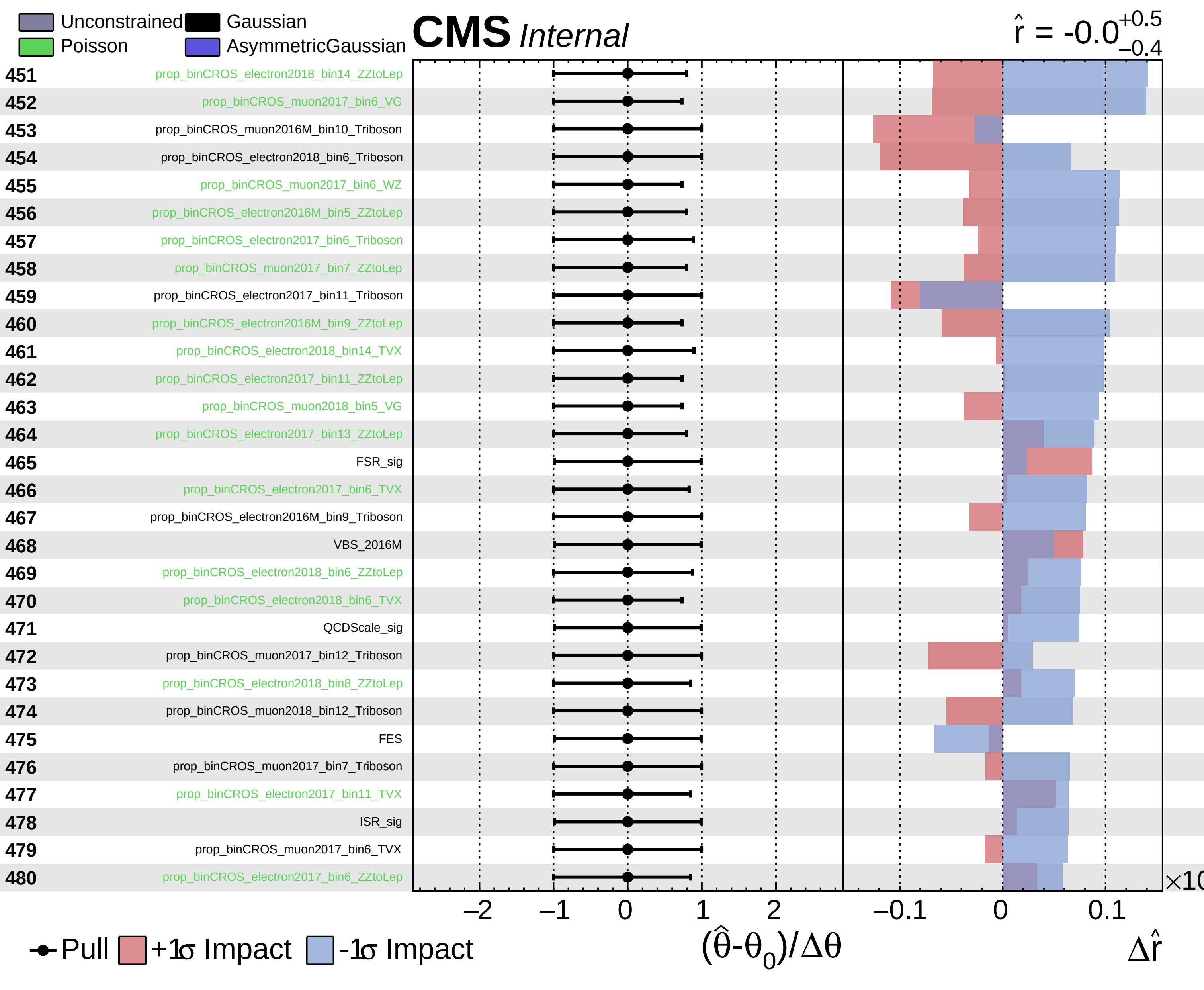
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.0^{+0.5}_{-0.4}$
331 prop_binCRTT_muon2017_bin13
332 prop_binCRTT_electron2018_bin5
333 prop_binCROS_electron2017_bin13_Fake
334 prop_binCRTT_muon2018_bin15
335 prop_binCRTT_muon2018_bin1
336 prop_binCRTT_electron2016M_bin5
337 prop_binCRTT_muon2018_bin16
338 prop_binCRTT_electron2018_bin11
339 prop_binCROS_electron2018_bin8_Fake
340 prop_binCRTT_electron2017_bin13
341 prop_binCROS_muon2018_bin12_Fake
342 prop_binCRTT_muon2018_bin8
343 prop_binCRTT_muon2016M_bin8
344 prop_binCRTT_electron2018_bin13
345 prop_binCRTT_electron2016M_bin15
346 prop_binCRTT_electron2016M_bin16
347 prop_binCRTT_muon2016M_bin5
348 prop_binCROS_electron2016M_bin13
349 prop_binCRTT_muon2016M_bin14
350 prop_binCROS_electron2018_bin8_WrongSign
351 prop_binCRTT_muon2017_bin3
352 prop_binCRTT_muon2017_bin4
353 prop_binCRTT_muon2017_bin2
354 prop_binCRTT_muon2016M_bin15
355 prop_binCRTT_muon2016M_bin12
356 prop_binCRTT_muon2016M_bin13
357 prop_binCRTT_electron2018_bin14
358 prop_binCROS_muon2017_bin15
359 prop_binCRTT_electron2017_bin7
360 prop_binCRTT_electron2016M_bin7
−2
−1
0
1
2
−0.2
0
0.2
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



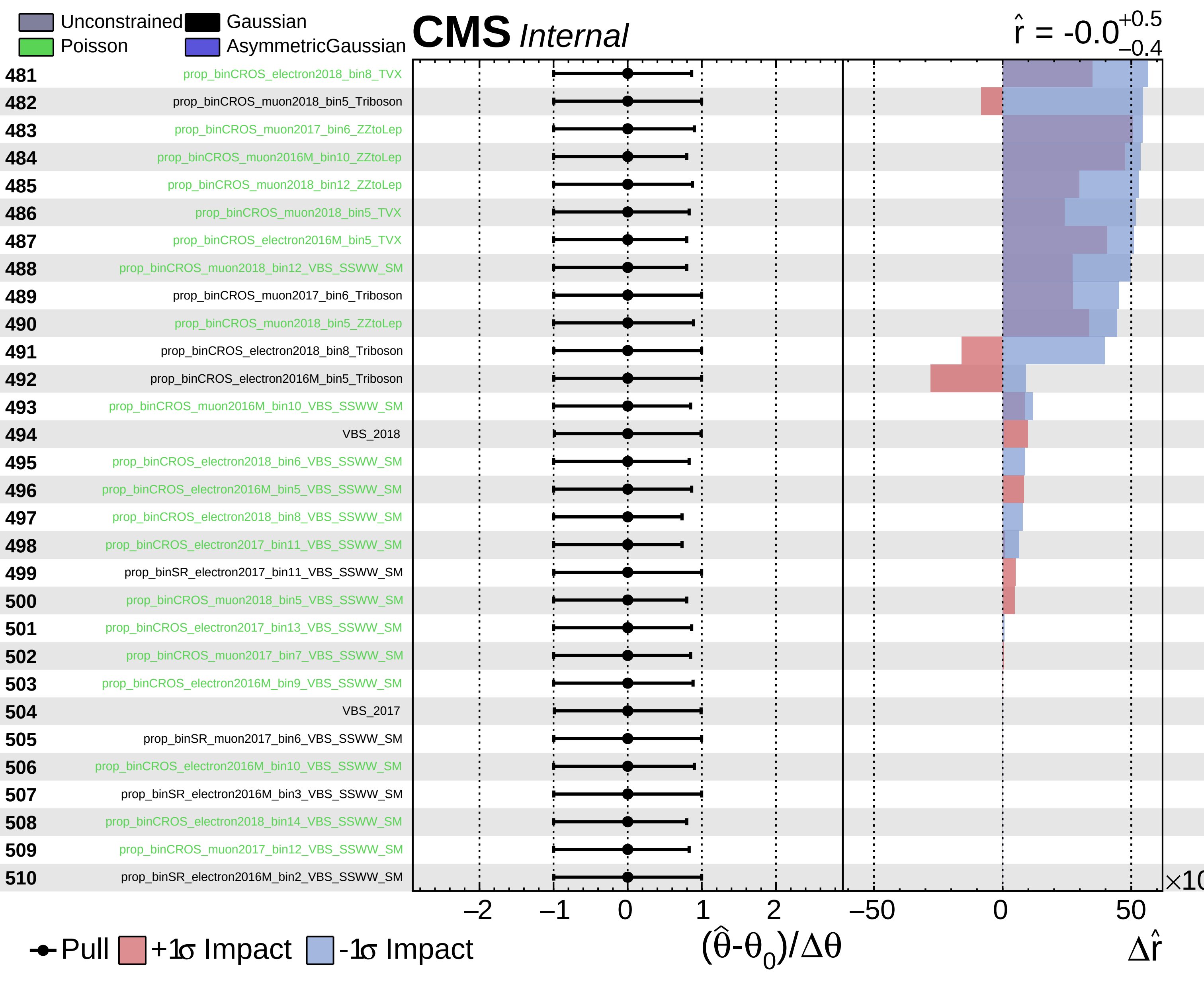
Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.0^{+0.5}_{-0.4}$
361 prop_binCRTT_electron2017_bin16
362 prop_binCRTT_electron2016M_bin13
363 prop_binCROS_muon2017_bin12_Fake
364 prop_binCRTT_electron2018_bin16
365 prop_binSR_electron2016M_bin2_WZ
366 prop_binCROS_electron2016M_bin16
367 prop_binSR_muon2017_bin6_TTTo2L2Nu
368 prop_binCROS_electron2018_bin14_TTTo2L2Nu
369 prop_binCROS_muon2017_bin7_Fake
370 prop_binCRTT_electron2017_bin4
371 prop_binCROS_electron2016M_bin6
372 prop_binCRTT_electron2016M_bin12
373 prop_binCRTT_electron2017_bin5
374 prop_binCRTT_muon2017_bin10
375 prop_binCROS_muon2016M_bin15
376 prop_binCRTT_electron2017_bin15
377 prop_binCRTT_electron2018_bin12
378 prop_binCRTT_muon2017_bin12
379 prop_binCRTT_electron2018_bin8
380 prop_binSR_electron2016M_bin3_VG
381 prop_binCROS_electron2016M_bin15
382 prop_binCRTT_electron2016M_bin10
383 prop_binCRTT_muon2017_bin11
384 prop_binCROS_electron2016M_bin10_Fake
385 prop_binCROS_muon2018_bin13
386 prop_binCRTT_muon2016M_bin16
387 prop_binSR_electron2016M_bin2_TTTo2L2Nu
388 prop_binSR_electron2016M_bin3_TTTo2L2Nu
389 prop_binCRTT_electron2017_bin3
390 prop_binCRTT_electron2017_bin14
−2 −1 0 1 2
−0.2 −0.1 0 0.1 0.2
×10
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$

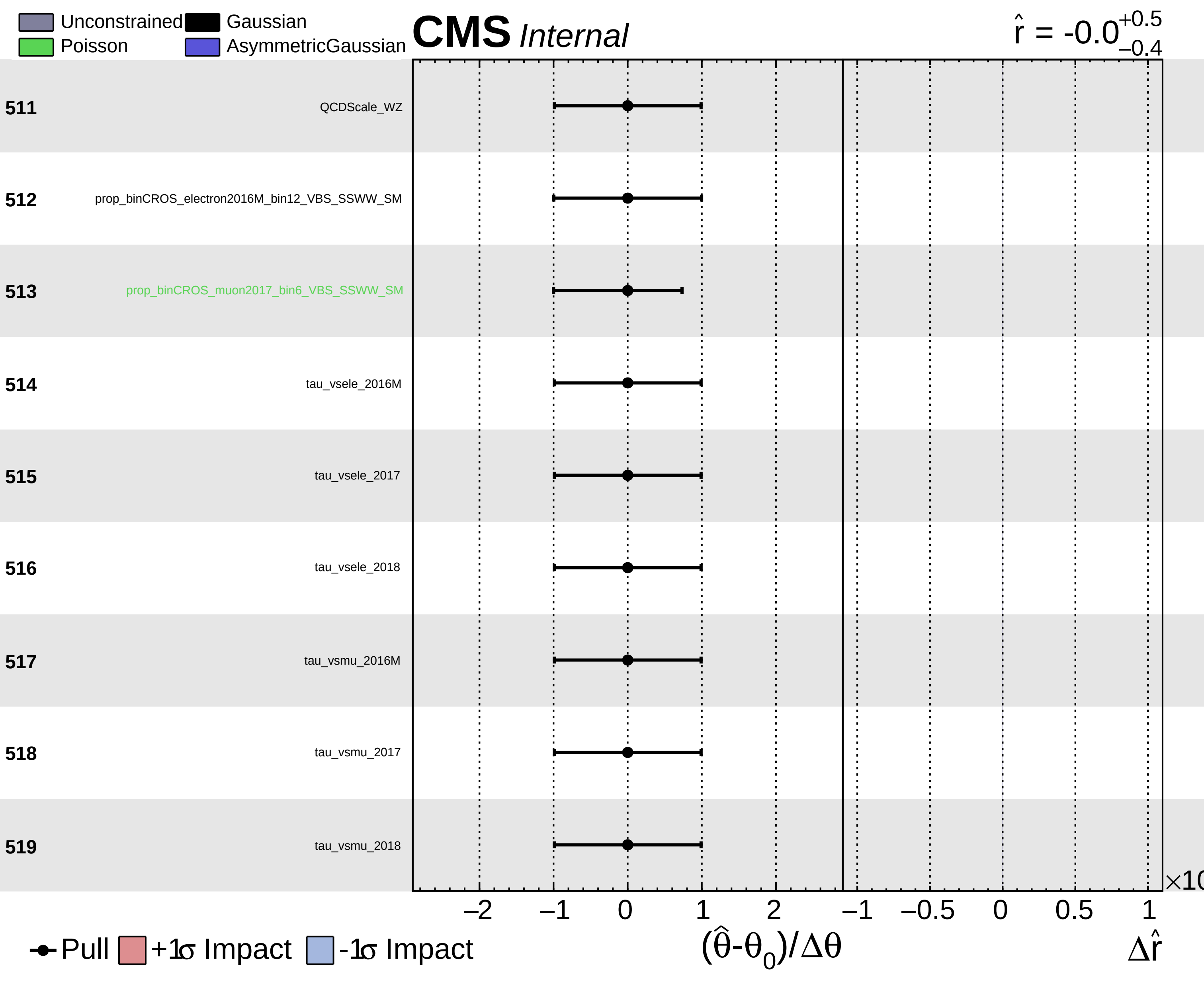

Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.0^{+0.5}_{-0.4}$
391 prop_binCRTT_electron2016M_bin9
392 prop_binCRTT_electron2018_bin6
393 prop_binCROS_muon2018_bin12_TTTo2L2Nu
394 prop_binCROS_muon2018_bin12_WrongSign
395 prop_binCRTT_electron2016M_bin14
396 prop_binCROS_muon2017_bin12_TTTo2L2Nu
397 prop_binCROS_muon2016M_bin10_Fake
398 prop_binCROS_electron2017_bin6_TTTo2L2Nu
399 prop_binCROS_muon2017_bin6_TTTo2L2Nu
400 prop_binCROS_electron2017_bin6_WZ
401 prop_binSR_electron2016M_bin3_ZZtoLep
402 lep_2016M
403 prop_binCROS_muon2016M_bin10_TTTo2L2Nu
404 prop_binCROS_electron2016M_bin10_TTTo2L2Nu
405 prop_binSR_electron2016M_bin2_VG
406 prop_binCROS_electron2016M_bin12_TTTo2L2Nu
407 prop_binCROS_muon2017_bin7_TTTo2L2Nu
408 prop_binCROS_electron2016M_bin5_TTTo2L2Nu
409 prop_binCROS_electron2016M_bin9_TTTo2L2Nu
410 prop_binSR_electron2016M_bin2_ZZtoLep
411 prop_binCROS_electron2017_bin14
412 prop_binCROS_muon2018_bin5_TTTo2L2Nu
413 prop_binCROS_electron2018_bin14_VG
414 prop_binSR_muon2017_bin6_WpWpJJ_QCD
415 prop_binSR_muon2017_bin6_TVX
416 prop_binSR_electron2016M_bin3_TVX
417 prop_binCROS_electron2017_bin13_TTTo2L2Nu
418 prop_binCROS_electron2018_bin8_TTTo2L2Nu
419 prop_binSR_muon2017_bin6_Triboson
420 lep_2018
−2 −1 0 1 2
−0.1 0 0.1
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.0^{+0.5}_{-0.4}$
421 prop_binCROS_electron2017_bin13_WZ
422 prop_binCROS_electron2018_bin8_VG
423 prop_binSR_electron2016M_bin3_WpWpJJ_QCD
424 prop_binSR_electron2016M_bin2_WpWpJJ_QCD
425 prop_binCROS_muon2018_bin12_TVX
426 prop_binCROS_muon2017_bin7_VG
427 prop_binSR_electron2016M_bin2_TVX
428 prop_binCROS_electron2018_bin6_TTTo2L2Nu
429 prop_binCROS_electron2017_bin11_TTTo2L2Nu
430 prop_binCROS_electron2016M_bin9_TVX
431 prop_binSR_electron2016M_bin3_Triboson
432 prop_binSR_electron2016M_bin2_Triboson
433 prop_binSR_muon2017_bin6_ZZtoLep
434 prop_binCROS_electron2016M_bin12_Fake
435 prop_binCROS_electron2017_bin13_TVX
436 prop_binCROS_electron2016M_bin10_VG
437 prop_binCROS_electron2016M_bin5_WZ
438 prop_binCROS_muon2017_bin12_TVX
439 prop_binCROS_electron2016M_bin10_TVX
440 prop_binCROS_electron2016M_bin12_Triboson
441 prop_binCROS_electron2017_bin13_Triboson
442 prop_binCROS_electron2016M_bin12_TVX
443 prop_binCROS_muon2017_bin7_TVX
444 prop_binCROS_muon2016M_bin10_VG
445 prop_binCROS_muon2016M_bin10_TVX
446 prop_binCROS_electron2016M_bin12_ZZtoLep
447 prop_binCROS_electron2016M_bin10_ZZtoLep
448 prop_binCROS_electron2016M_bin10_Triboson
449 prop_binCROS_muon2017_bin12_ZZtoLep
450 prop_binCROS_electron2018_bin14_Triboson
−2
−1
0
1
2
−0.01
0
0.01
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.0^{+0.5}_{-0.4}$
451 prop_binCROS_electron2018_bin14_ZZtoLep
452 prop_binCROS_muon2017_bin6_VG
453 prop_binCROS_muon2016M_bin10_Triboson
454 prop_binCROS_electron2018_bin6_Triboson
455 prop_binCROS_muon2017_bin6_WZ
456 prop_binCROS_electron2016M_bin5_ZZtoLep
457 prop_binCROS_electron2017_bin6_Triboson
458 prop_binCROS_muon2017_bin7_ZZtoLep
459 prop_binCROS_electron2017_bin11_Triboson
460 prop_binCROS_electron2016M_bin9_ZZtoLep
461 prop_binCROS_electron2018_bin14_TVX
462 prop_binCROS_electron2017_bin11_ZZtoLep
463 prop_binCROS_muon2018_bin5_VG
464 prop_binCROS_electron2017_bin13_ZZtoLep
465 FSR_sig
466 prop_binCROS_electron2017_bin6_TVX
467 prop_binCROS_electron2016M_bin9_Triboson
468 VBS_2016M
469 prop_binCROS_electron2018_bin6_ZZtoLep
470 prop_binCROS_electron2018_bin6_TVX
471 QCDScale_sig
472 prop_binCROS_muon2017_bin12_Triboson
473 prop_binCROS_electron2018_bin8_ZZtoLep
474 prop_binCROS_muon2018_bin12_Triboson
475 FES
476 prop_binCROS_muon2017_bin7_Triboson
477 prop_binCROS_electron2017_bin11_TVX
478 ISR_sig
479 prop_binCROS_muon2017_bin6_TVX
480 prop_binCROS_electron2017_bin6_ZZtoLep
−2 −1 0 1 2
−0.1 0 0.1
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$


Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.0^{+0.5}_{-0.4}$
481 prop_binCROS_electron2018_bin8_TVX
482 prop_binCROS_muon2018_bin5_Triboson
483 prop_binCROS_muon2017_bin6_ZZtoLep
484 prop_binCROS_muon2016M_bin10_ZZtoLep
485 prop_binCROS_muon2018_bin12_ZZtoLep
486 prop_binCROS_muon2018_bin5_TVX
487 prop_binCROS_electron2016M_bin5_TVX
488 prop_binCROS_muon2018_bin12_VBS_SSWW_SM
489 prop_binCROS_muon2017_bin6_Triboson
490 prop_binCROS_muon2018_bin5_ZZtoLep
491 prop_binCROS_electron2018_bin8_Triboson
492 prop_binCROS_electron2016M_bin5_Triboson
493 prop_binCROS_muon2016M_bin10_VBS_SSWW_SM
494 VBS_2018
495 prop_binCROS_electron2018_bin6_VBS_SSWW_SM
496 prop_binCROS_electron2016M_bin5_VBS_SSWW_SM
497 prop_binCROS_electron2018_bin8_VBS_SSWW_SM
498 prop_binCROS_electron2017_bin11_VBS_SSWW_SM
499 prop_binSR_electron2017_bin11_VBS_SSWW_SM
500 prop_binCROS_muon2018_bin5_VBS_SSWW_SM
501 prop_binCROS_electron2017_bin13_VBS_SSWW_SM
502 prop_binCROS_muon2017_bin7_VBS_SSWW_SM
503 prop_binCROS_electron2016M_bin9_VBS_SSWW_SM
504 VBS_2017
505 prop_binSR_muon2017_bin6_VBS_SSWW_SM
506 prop_binCROS_electron2016M_bin10_VBS_SSWW_SM
507 prop_binSR_electron2016M_bin3_VBS_SSWW_SM
508 prop_binCROS_electron2018_bin14_VBS_SSWW_SM
509 prop_binCROS_muon2017_bin12_VBS_SSWW_SM
510 prop_binSR_electron2016M_bin2_VBS_SSWW_SM
−2
−1
0
1
2
−50
0
50
×10
Pull
+1σ Impact
-1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$



Unconstrained
Gaussian
Poisson
AsymmetricGaussian
CMS Internal
$\hat{r} = -0.0^{+0.5}_{-0.4}$
511 QCDScale_WZ
512 prop_binCROS_electron2016M_bin12_VBS_SSWW_SM
513 prop_binCROS_muon2017_bin6_VBS_SSWW_SM
514 tau_vsele_2016M
515 tau_vsele_2017
516 tau_vsele_2018
517 tau_vsmu_2016M
518 tau_vsmu_2017
519 tau_vsmu_2018
−2 −1 0 1 2
−1 −0.5 0 0.5 1
×10
Pull +1σ Impact -1σ Impact
$(\hat{\theta}-\theta_0)/\Delta\theta$
$\Delta\hat{r}$